Report_DEB

1. Importation des données

```
file_path <- here::here("mod/L_DEB_SAAWKG2e")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 6333, C2 = 3665, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,           0.1,      0.2,  0.01,      0.8);
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,        0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,    0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,               0,     1);
Distrib (kapA,         Uniform,               0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,          30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,         0.00001,      0.3);
Distrib (K,            Uniform,            0.00001,        2);
Distrib (dv,           TruncNormal_cv,           2,      0.5,     0,     10);
Distrib (wE,           LogUniform,      0.000000001,     0.1);
Distrib (gOMsoil,          LogUniform,               0.0001, 1);
Distrib (gOMhorse,          LogUniform,               0.0001, 1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.51254800	137.88900	0.85092800	0.17492500	0.95469700	0.71290600
2.5%	0.40041400	100.42555	0.75718475	0.14669700	0.92472400	0.63112100
97.5%	0.62057700	167.62475	1.03455000	0.24594900	0.98037400	0.88012900
Min	0.15171600	85.69340	0.66854500	0.13350800	0.07193230	0.19675200
Max	0.76243700	182.90500	1.19691000	0.26550600	0.98554200	0.89991900
Result_mean	0.49722692	137.92681	0.89284309	0.18768271	0.95859973	0.74328772
Result_sd	0.05733262	17.35667	0.07273071	0.02479827	0.01619112	0.06461905

	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.
Result_mode	93.73950	25.585300	0.0039909500	0.39121500	1.988930
2.5%	73.18850	20.064100	0.0000169116	0.28273225	0.754145
97.5%	151.49500	40.266525	0.0075542265	0.57299800	2.404160
Min	64.09140	16.903100	0.0000105763	0.21220600	0.613017
Max	175.12800	50.926200	0.0708173000	0.91000600	3.167680
Result_mean	112.79897	29.105939	0.0029561392	0.45341374	1.577221
Result_sd	19.71279	5.247151	0.0024260158	0.07431319	0.419076

	wE.1.	gOMsoil.1.	gOMhorse.1.	Sigma_W.1.
Result_mode	1.170860e-05	0.000205979	0.000134120	0.10179100
2.5%	2.575386e-06	0.000119437	0.000120743	0.09508733
97.5%	1.621557e-05	0.448394150	0.141558150	0.12205770
Min	1.537300e-09	0.000110322	0.000100049	0.08422940
Max	1.769270e-05	0.557667000	0.198670000	5.07019000
Result_mean	1.198758e-05	0.118420473	0.028310495	0.10801728
Result_sd	2.671396e-06	0.155113203	0.039814778	0.04632501

Number of observations, n = 277
Number of parameters, k = 14
Log-likelihood, LL = 11.2004
BIC = 56.33545

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	2.57	5.20
E_coc.1.	1.06	1.20
Ehp.1.	1.15	1.46
Fm.1.	1.05	1.13
gOMhorse.1.	1.55	2.89
gOMsoil.1.	1.55	3.08

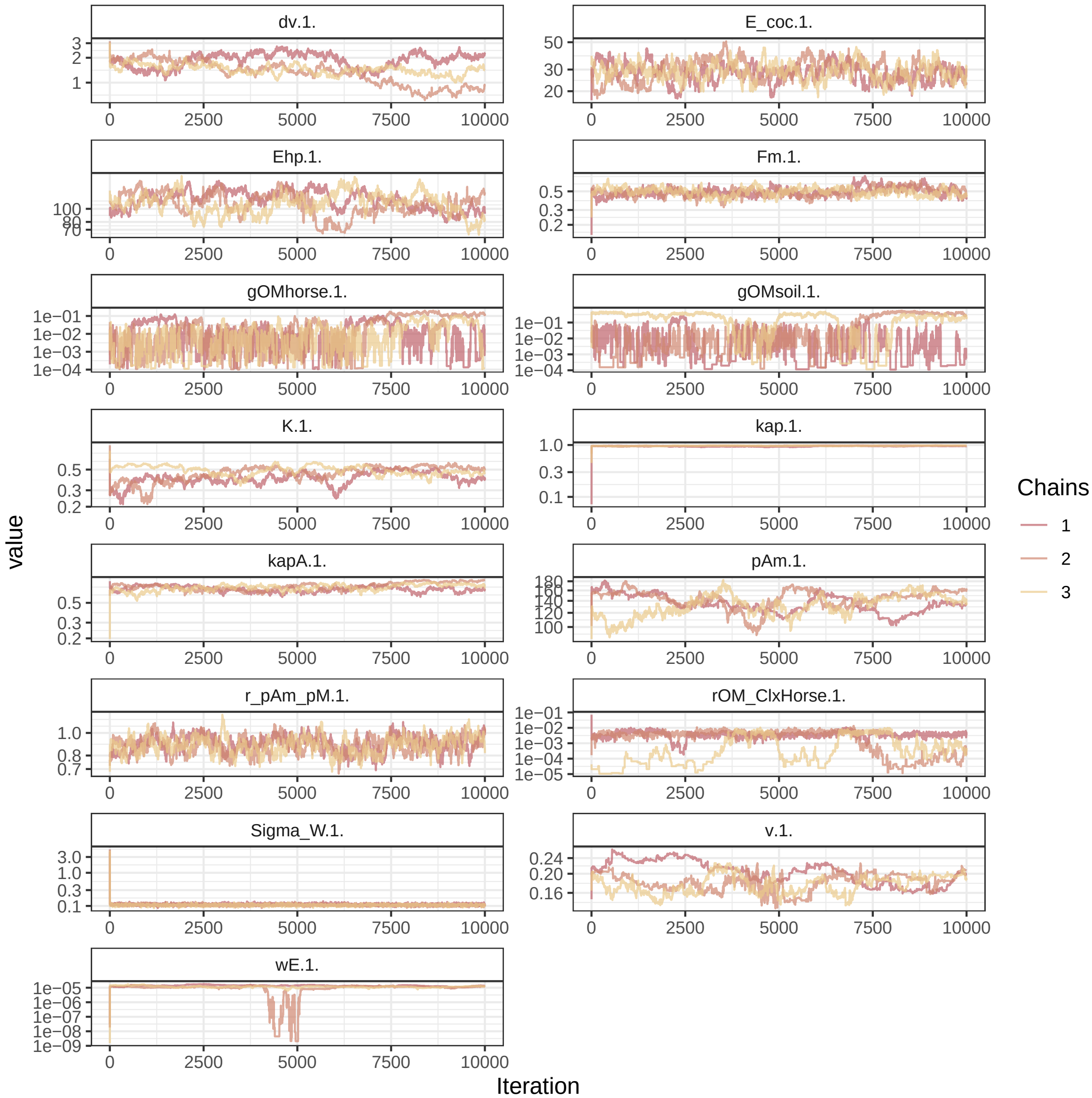


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

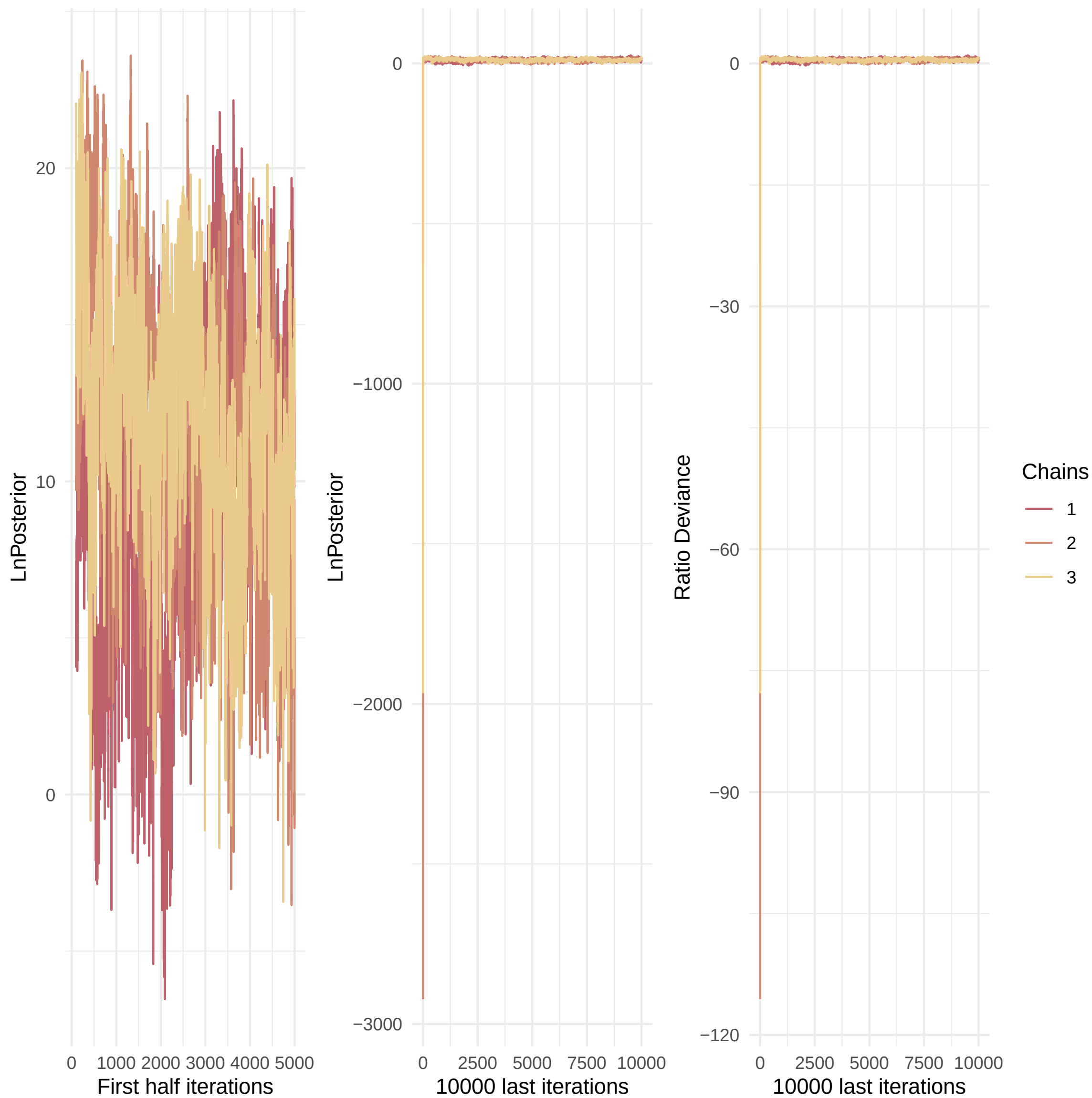


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
K.1.                  1.38       2.01
kap.1.                2.77       6.21
kapA.1.               1.77       2.92
pAm.1.                1.41       2.09
r_pAm_pM.1.           1.01       1.02
rOM_ClxHorse.1.       1.18       1.60
Sigma_W.1.            1.18       1.51
v.1.                  1.04       1.13
wE.1.                 1.47       2.59

Multivariate psrf

2.67
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

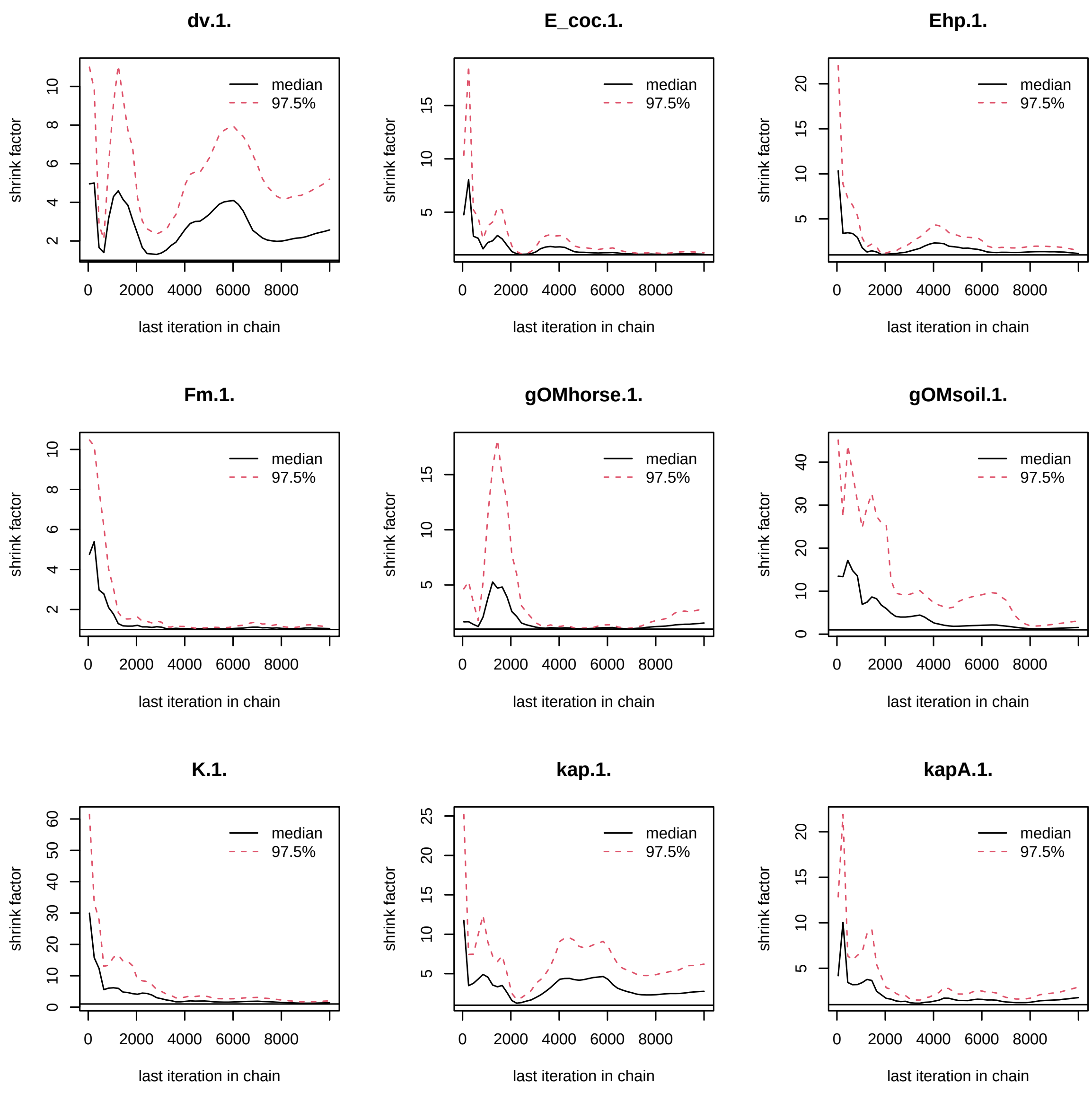


Figure 3: Chains convergence.

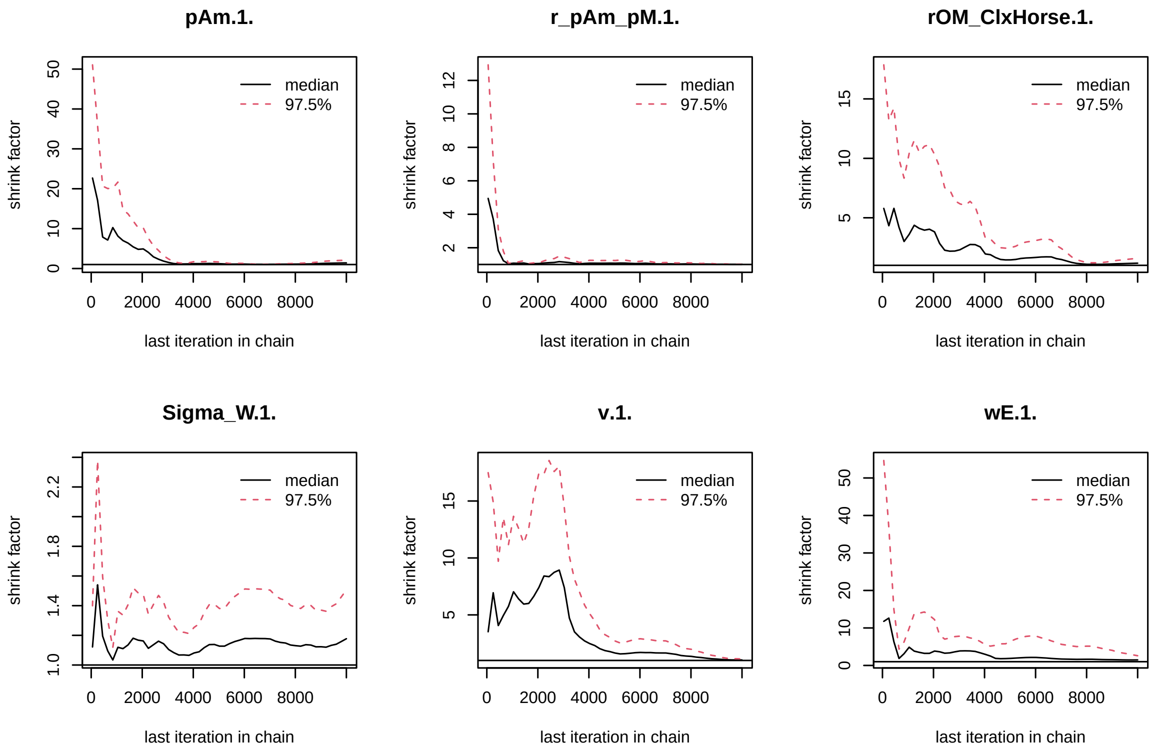


Figure 4: Chains convergence.

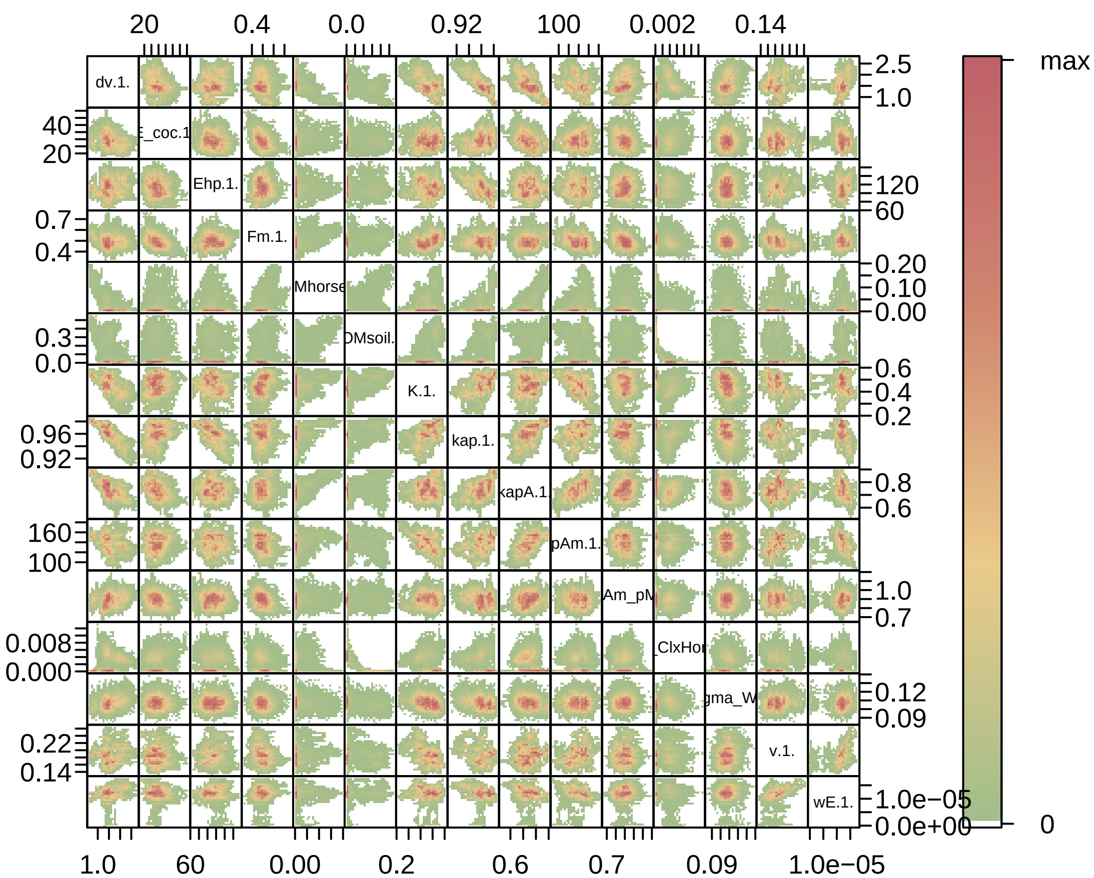


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

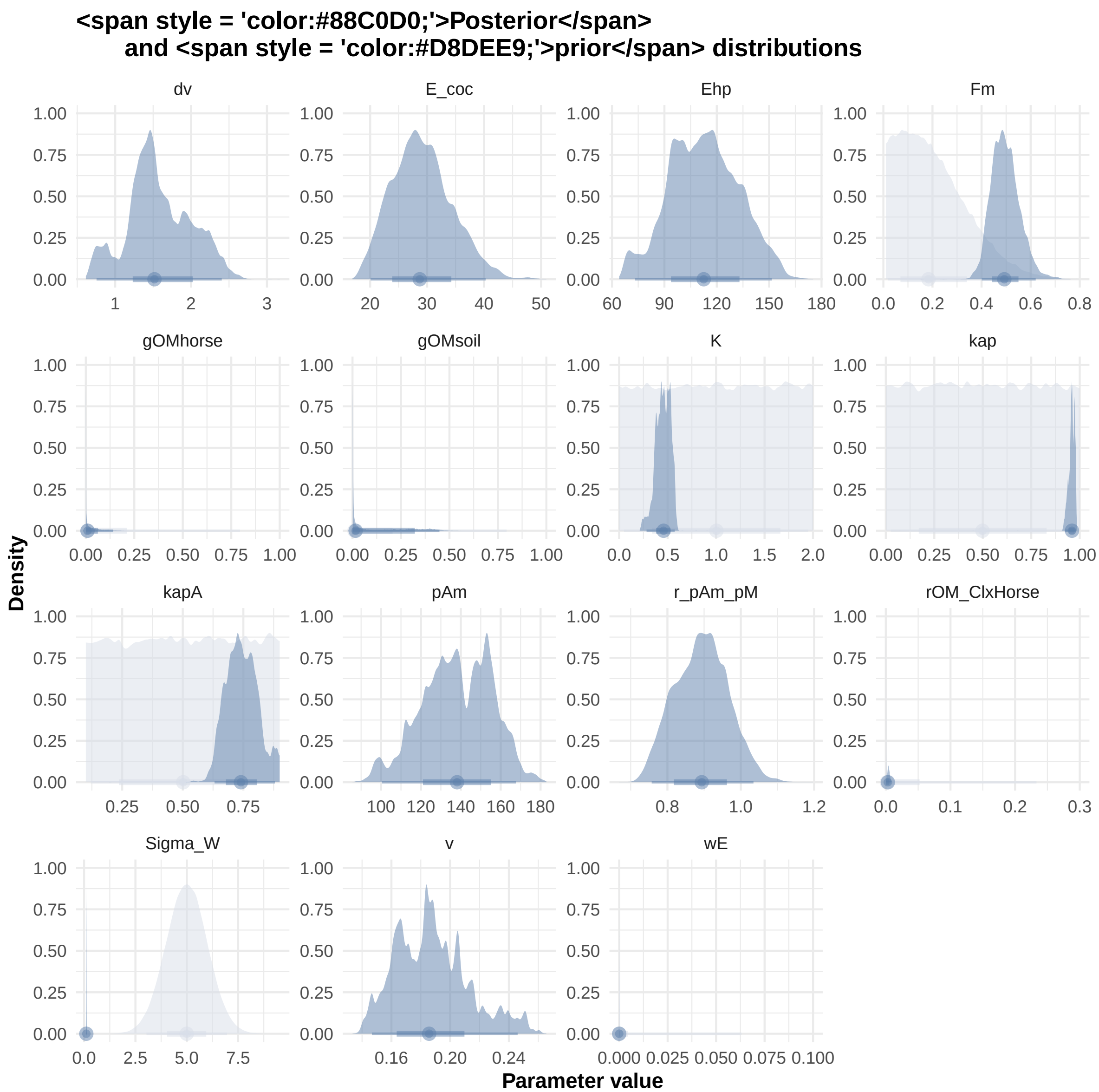


Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pri

# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
```

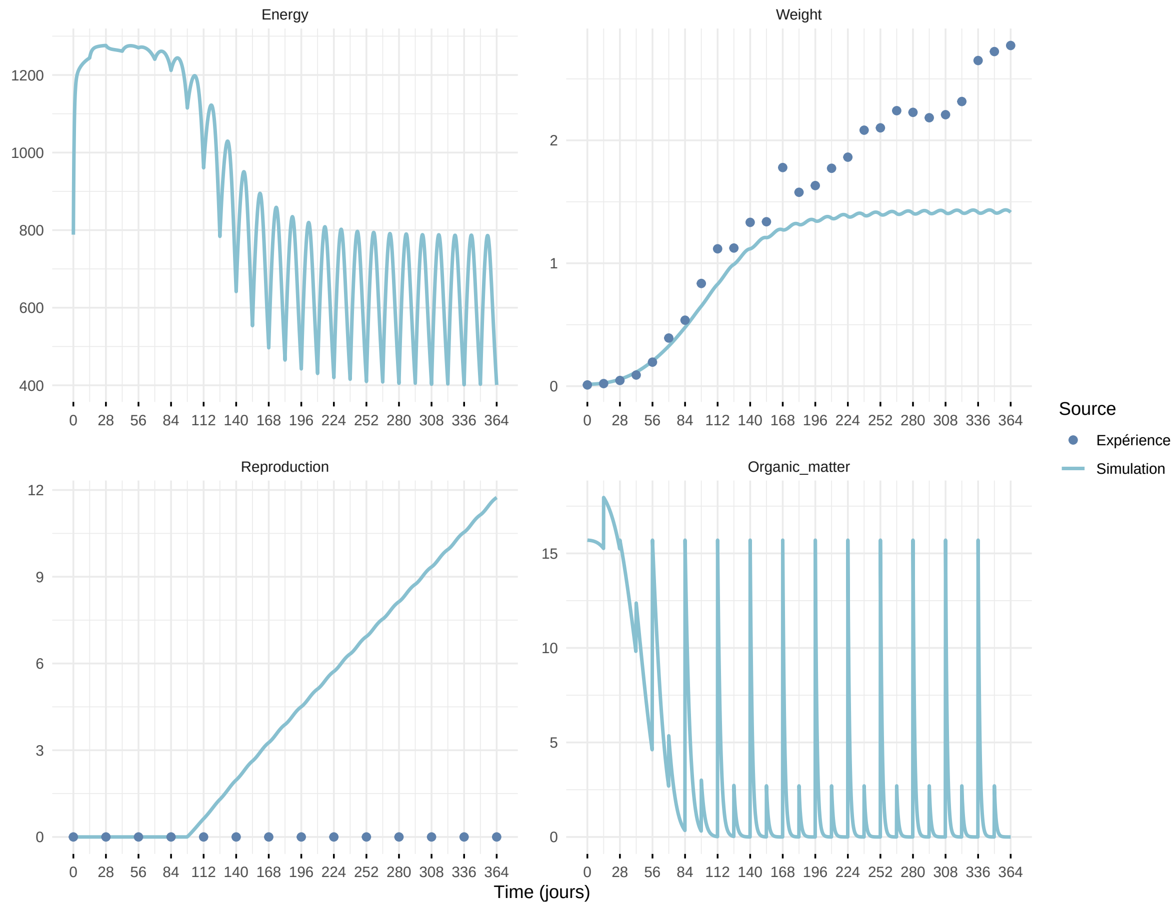
```r
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

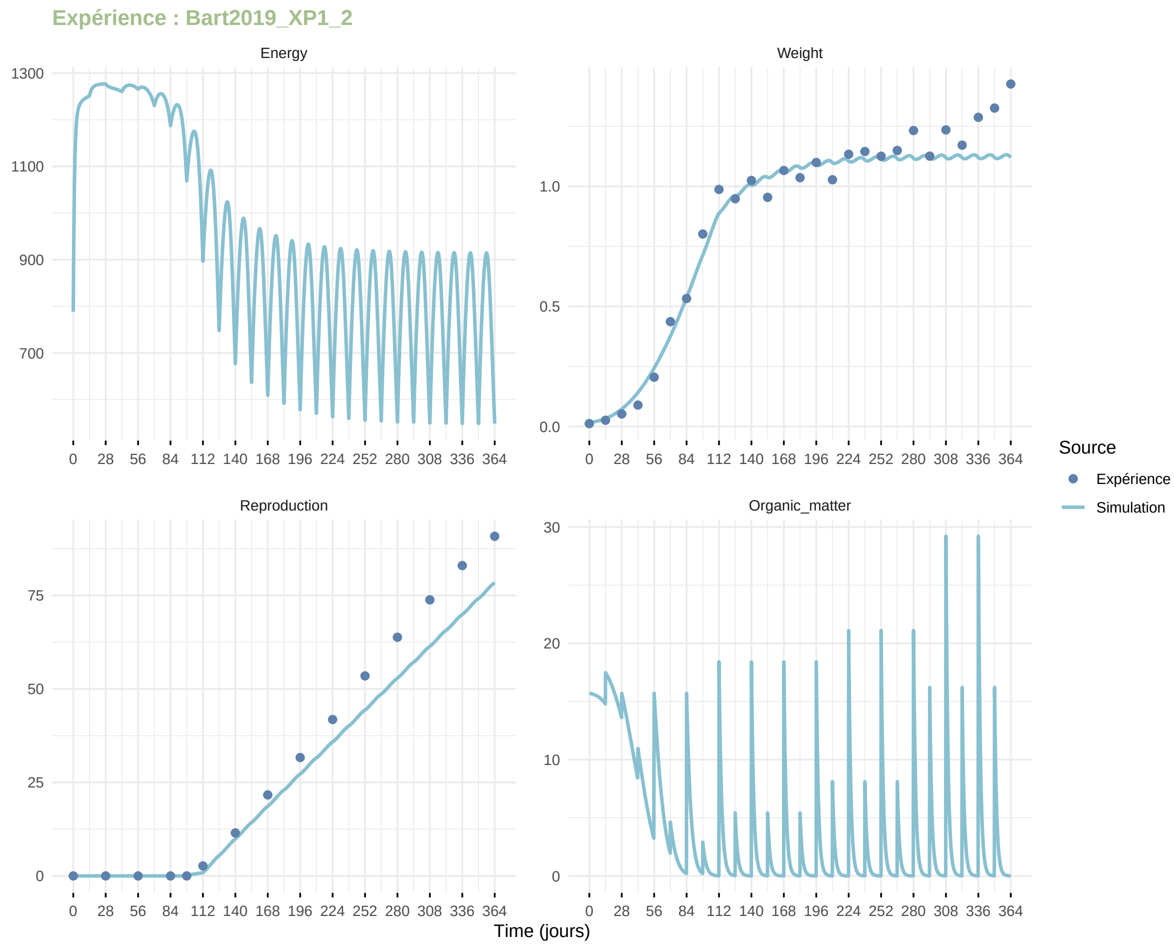
```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

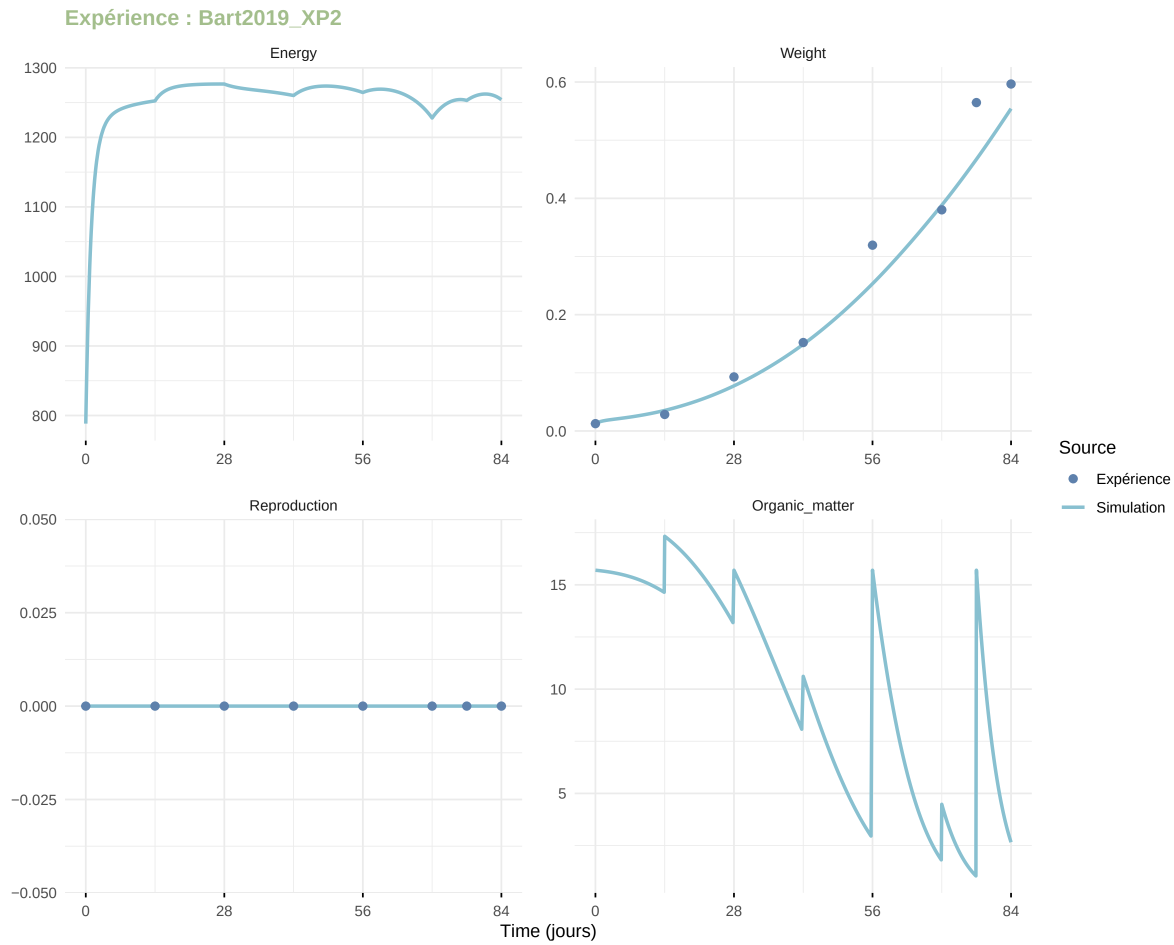
Figure 7: Simulations Weight and Reproduction.

Expérience : Bart2019_XP1_1

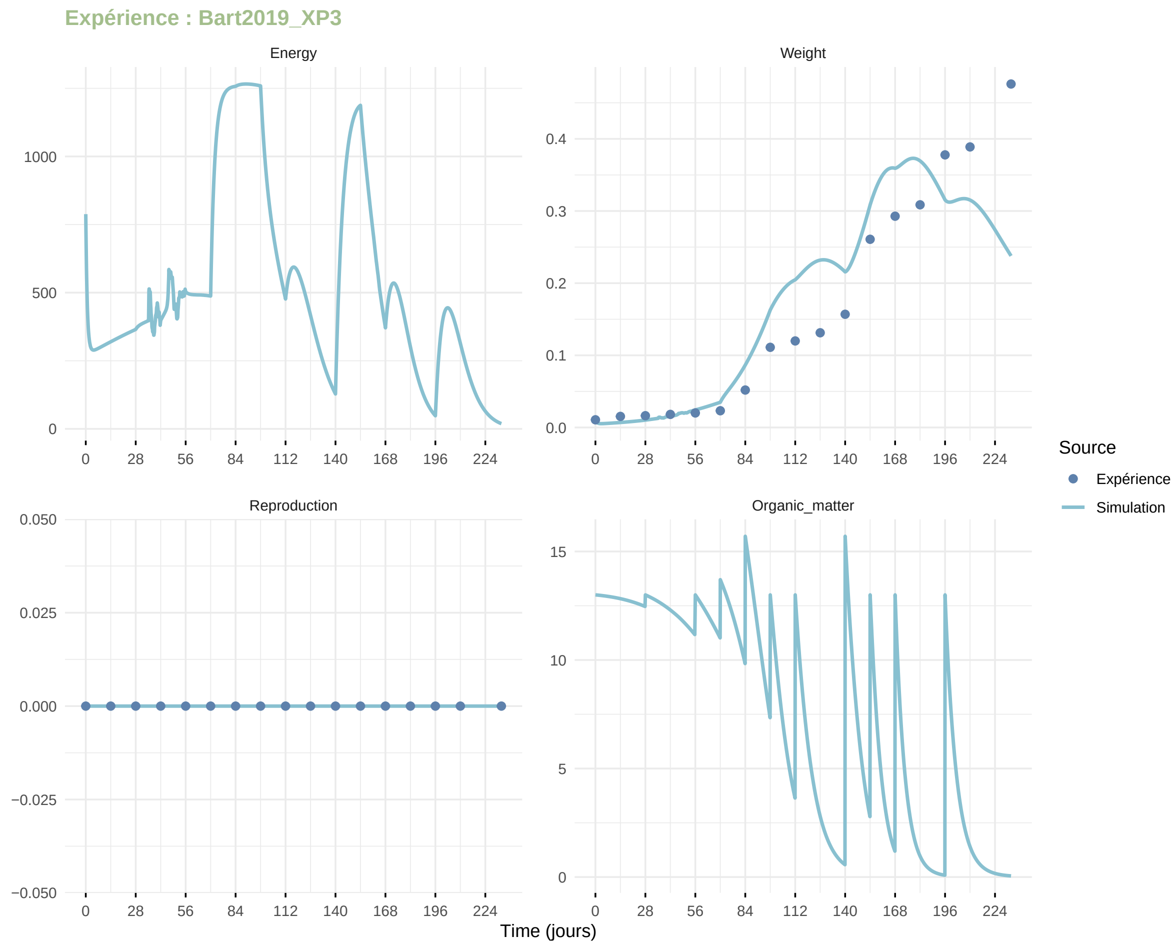




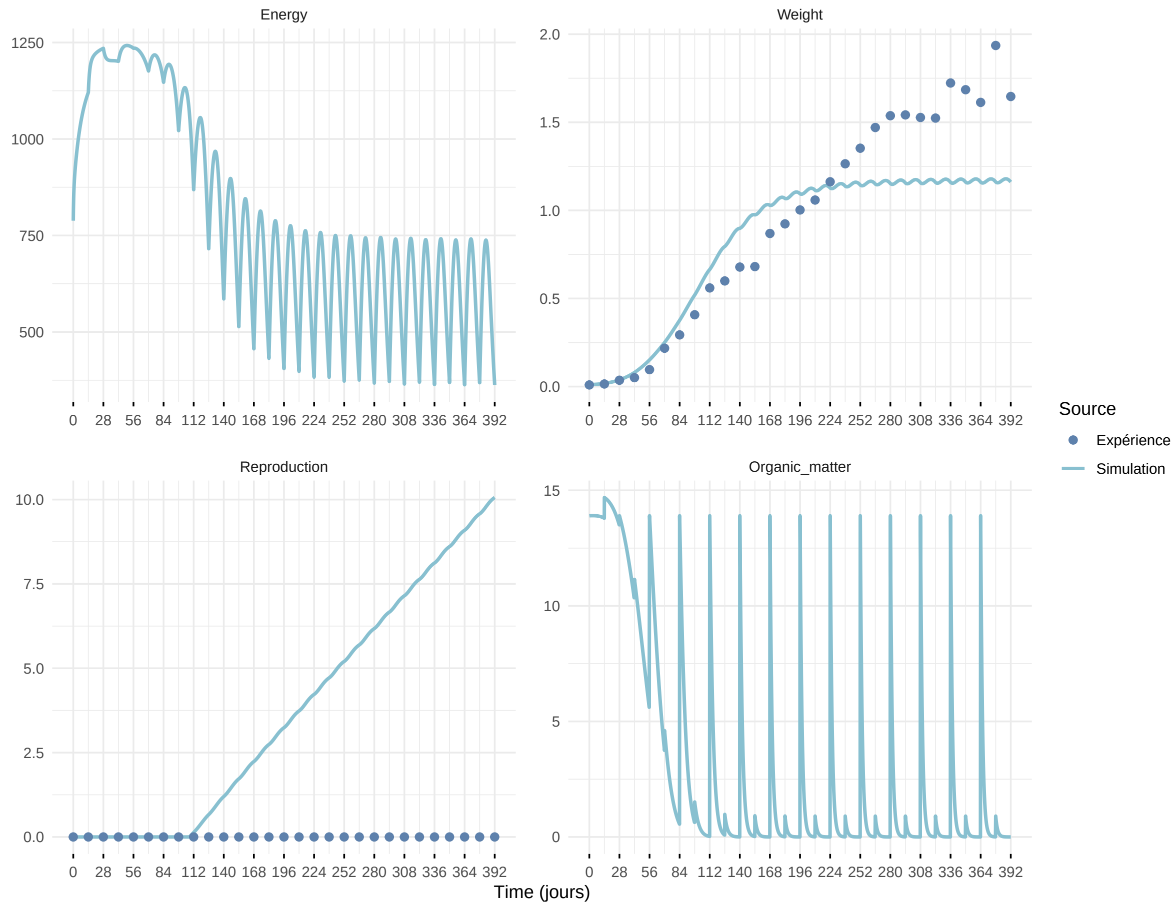
Expérience : Bart2019_XP1_2
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)



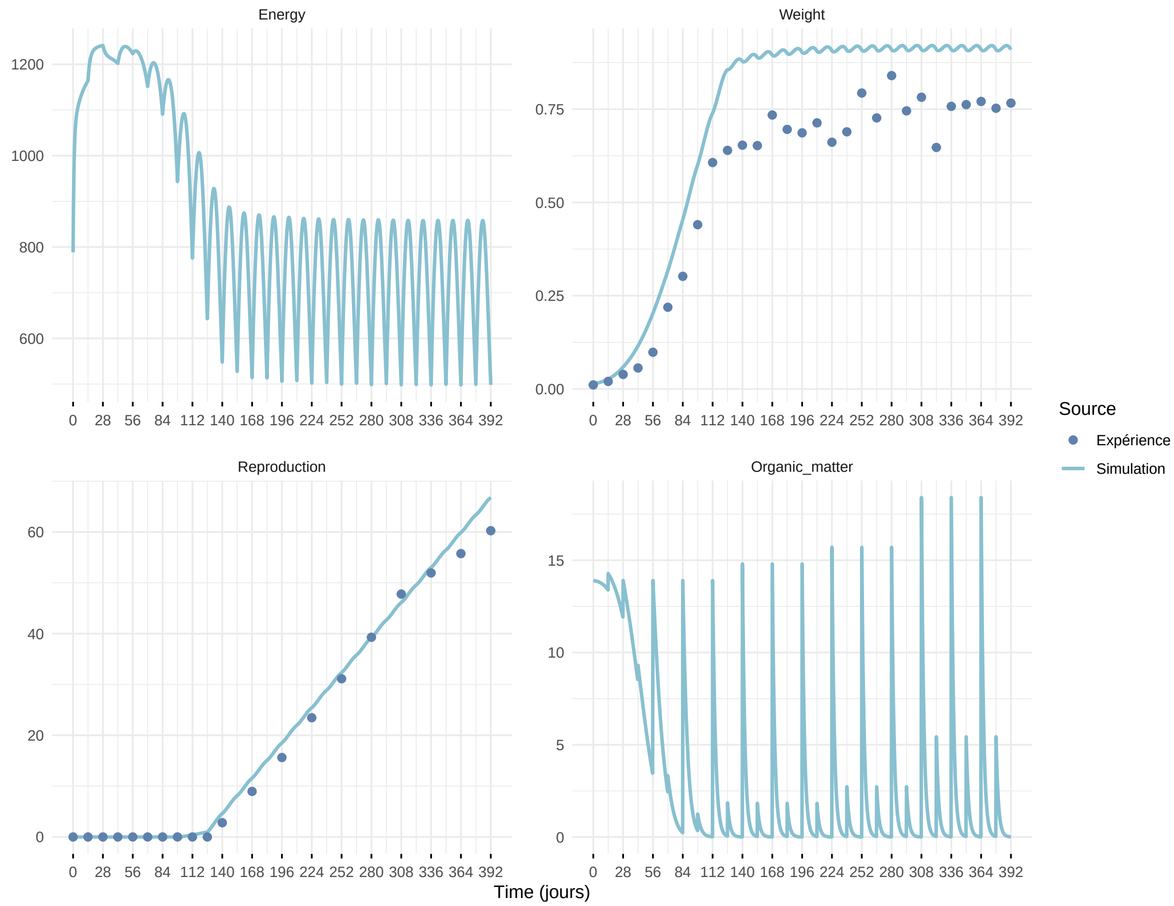
Expérience : Bart2019_XP2
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

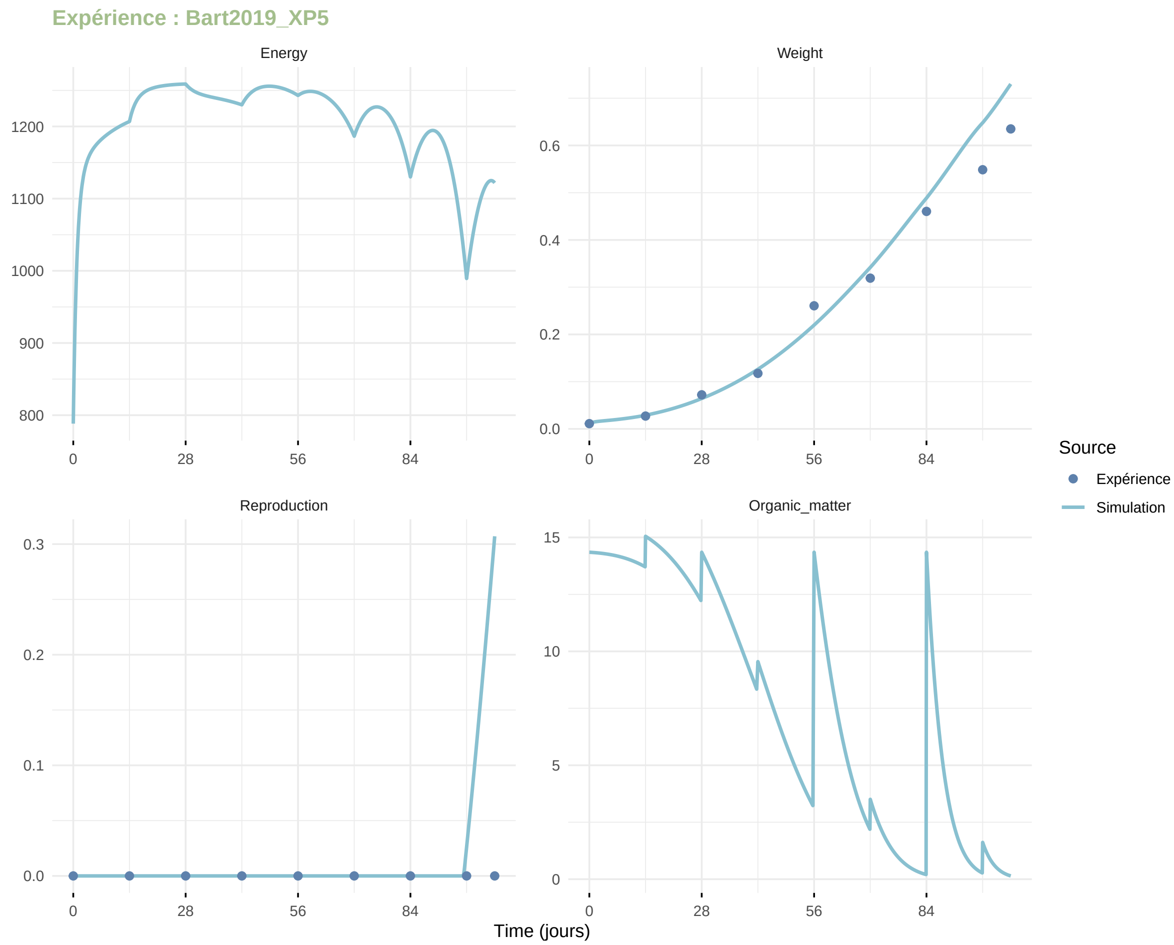

Expérience : Bart2019_XP3
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

Expérience : Bart2019_XP4_1

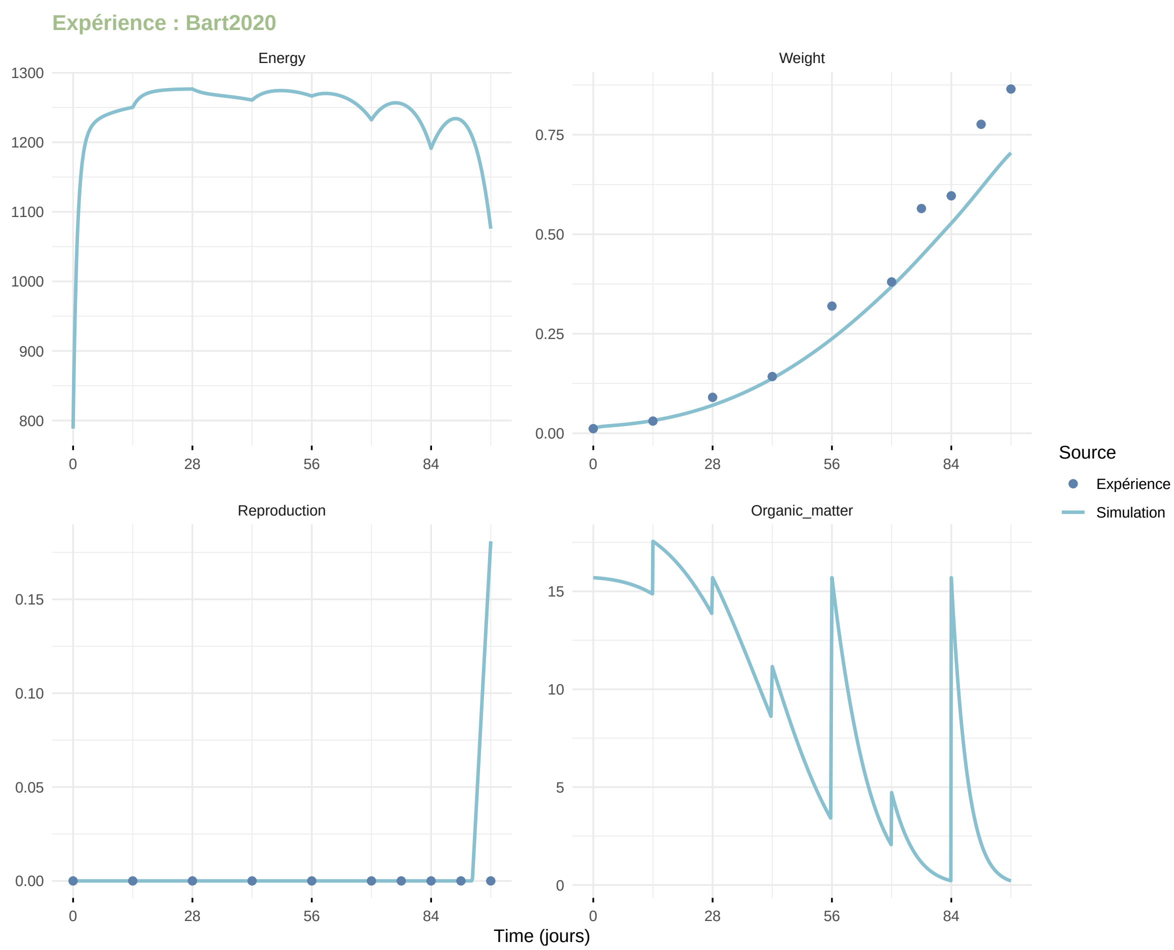


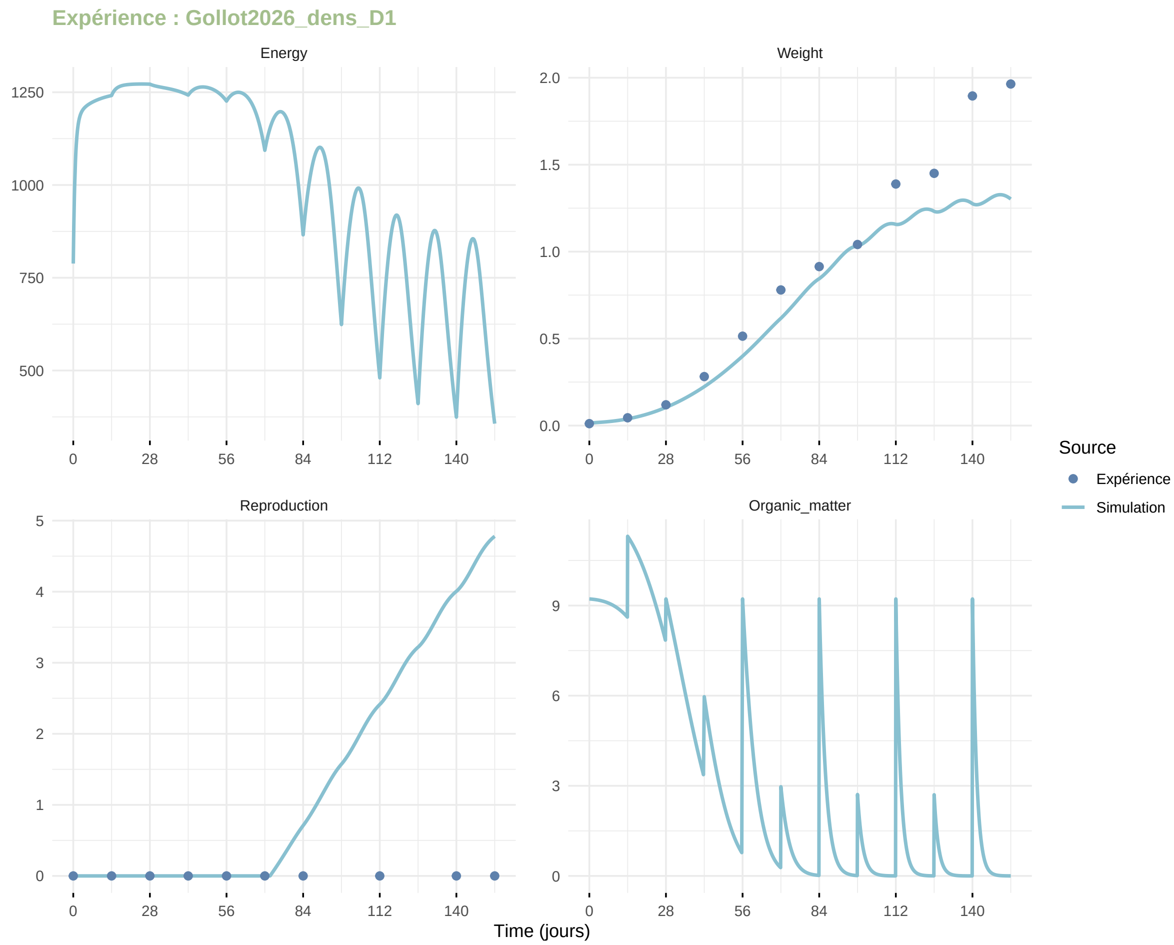
Expérience : Bart2019_XP4_2

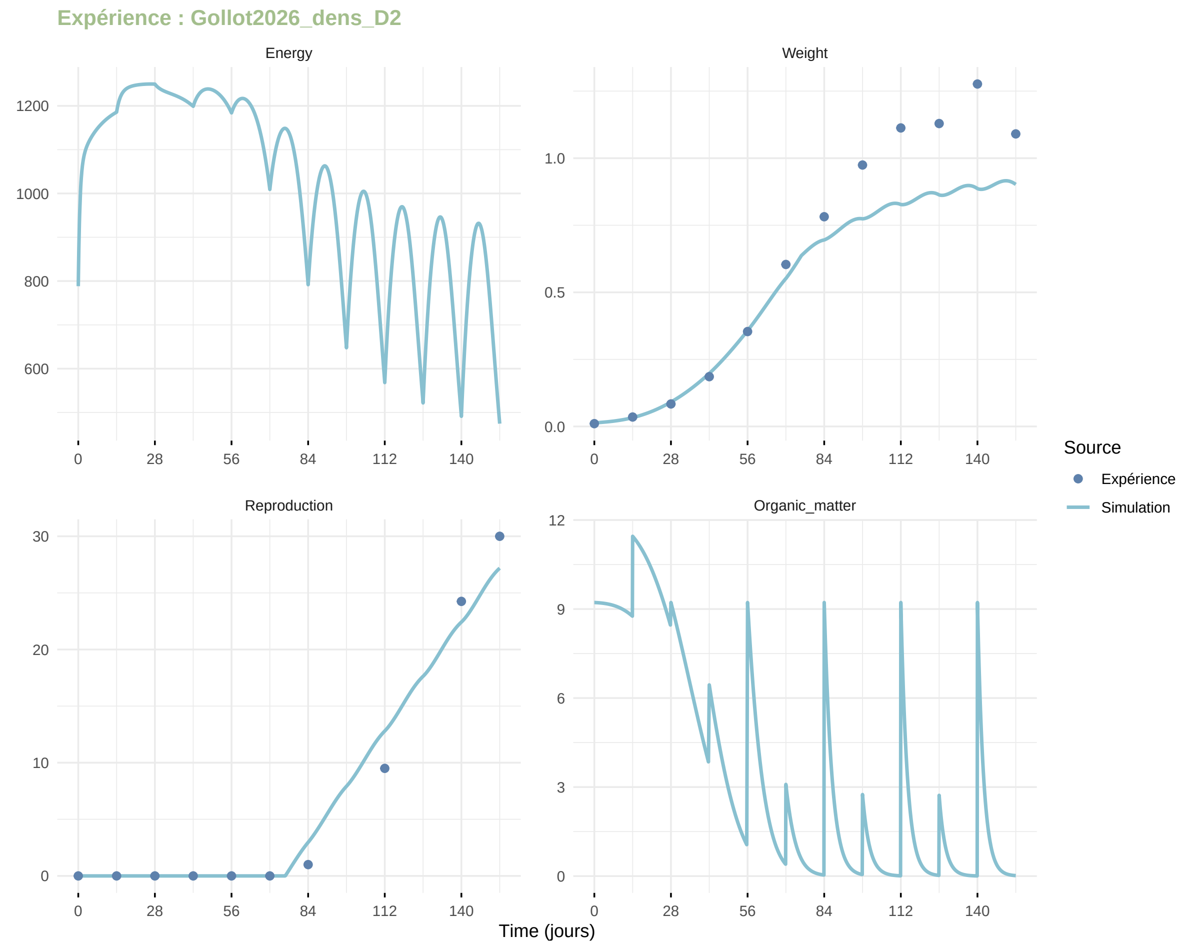




Expérience : Bart2019_XP5
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)
0
28
56
84

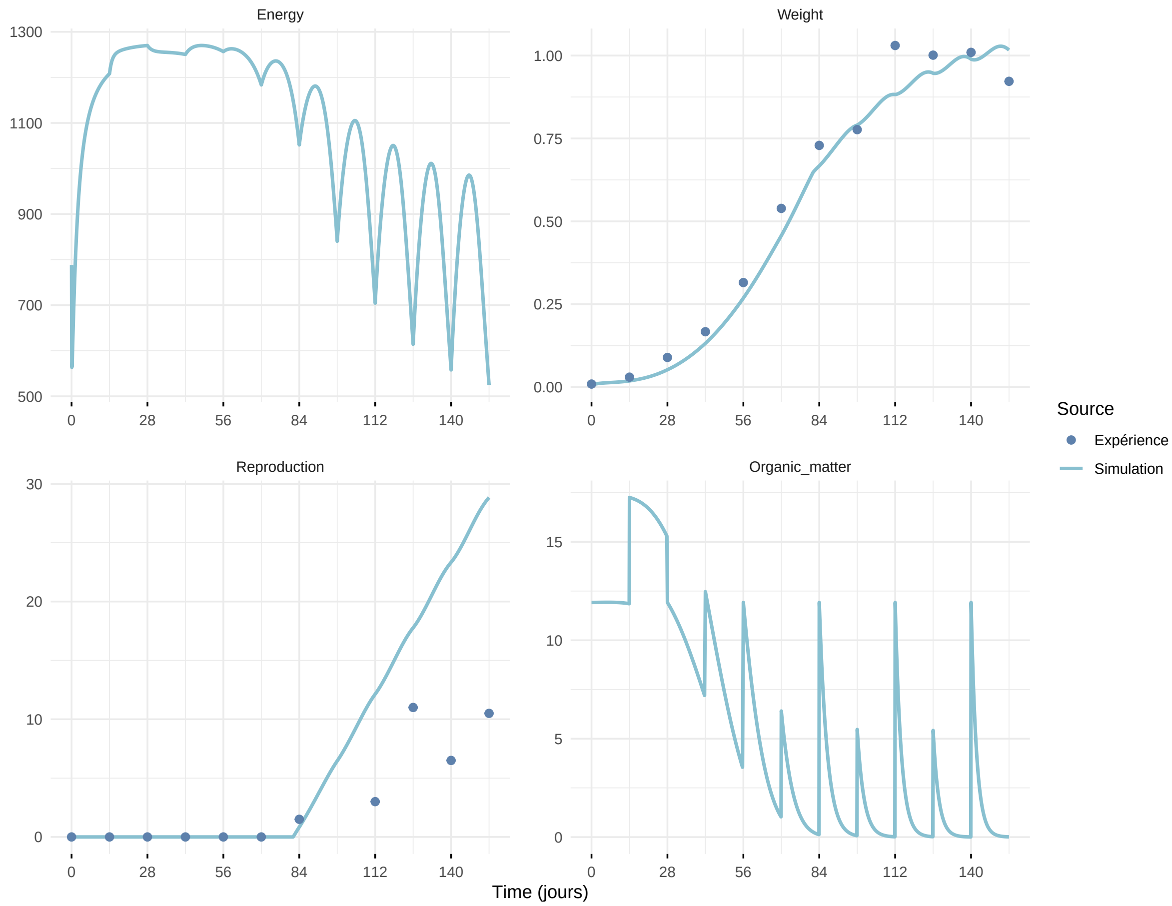

Expérience : Bart2020
Energy
1300
1200
1100
1000
900
800
0
28
56
84
Weight
0.75
0.50
0.25
0.00
0
28
56
84
Source
Expérience
Simulation
Reproduction
0.15
0.10
0.05
0.00
0
28
56
84
Organic_matter
15
10
5
0
0
28
56
84
Time (jours)


Expérience : Gollot2026_dens_D1
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

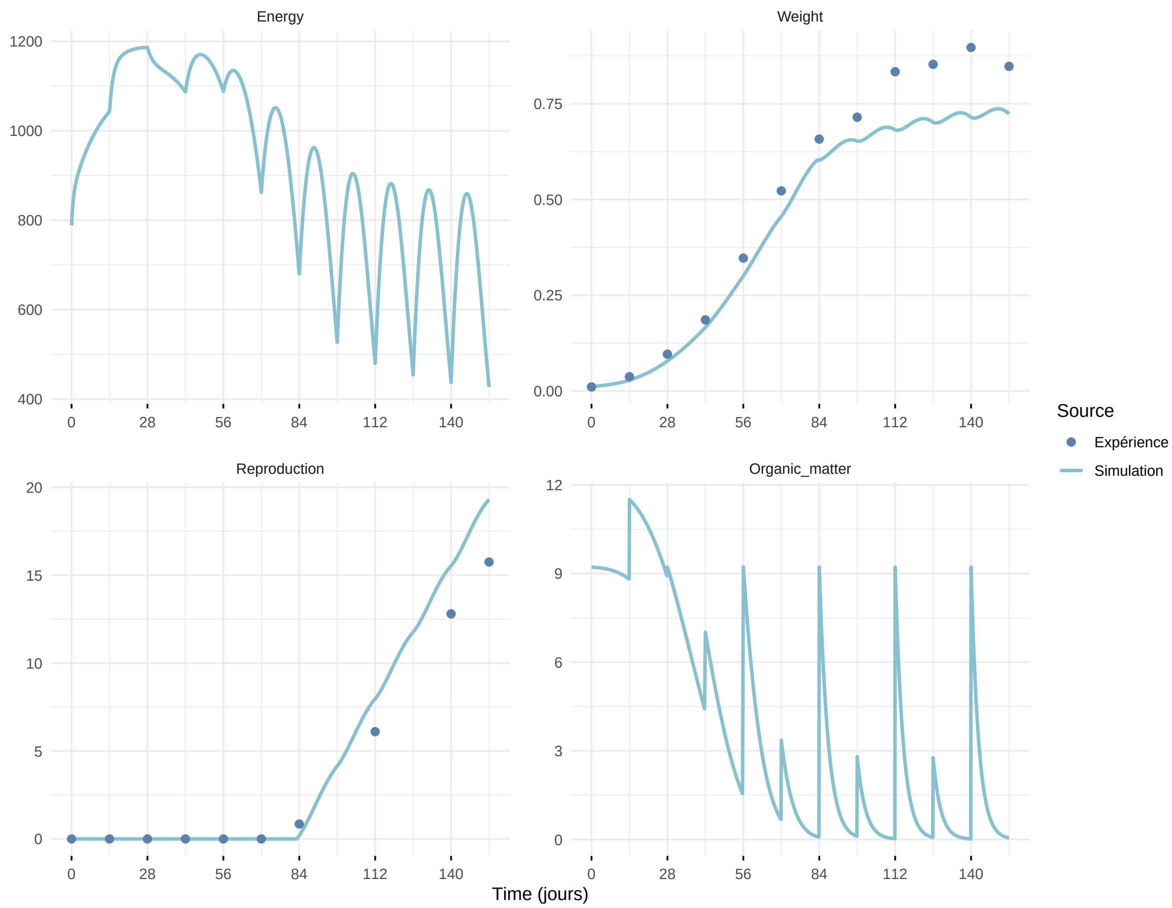


Expérience : Gollot2026_dens_D2
Energy
Weight
Reproduction
Organic_matter
1200
1000
800
600
1.0
0.5
0.0
30
20
10
0
12
9
6
3
0
28
56
84
112
140
Time (jours)
Source
Expérience
Simulation

Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7